Assignment Lab 2.2 — Supervised Learning for Cybersecurity Analysis

📘 Course 02 – Introduction to Machine Learning for Cybersecurity

🎯 Assignment Overview

In this assignment, you will independently apply supervised machine learning techniques to analyze a cybersecurity-related dataset. You will use Python and Scikit-learn to train a machine learning model, generate predictions, and evaluate the results.

This assignment builds on the concepts and workflow introduced in Guided Lab 2.1.

📂 Dataset

Use the provided dataset:

✅ `cybersecurity_login_dataset.csv`

The dataset contains login activity information that can be used to identify suspicious behavior.

📌 Important Notes

- Use the provided starter notebook or create your own notebook structure

Starter Notebook

- Remember to update the dataset path to match the location of your downloaded dataset
- Ensure your notebook runs without errors before submitting
- Clearly label all sections of your work

💡 Tips for Success

- Review Guided Lab 2.1 before starting
- Compare your predictions with actual outputs carefully
- Focus on understanding the workflow, not only the final accuracy score
- Use comments in your code to explain your steps

🛠 Assignment Tasks

Part 1 — Load and Explore the Dataset

- Import the dataset into Python
- Display the first few rows of the dataset
- Review dataset information and check for missing values

Part 2 — Prepare the Data

- Define the feature columns (X)
- Define the target variable (y)
- Split the data into training and testing datasets

Part 3 — Train a Supervised Learning Model

Choose and train at least **one** of the following models:

- Logistic Regression
- Decision Tree
- Random Forest
- Support Vector Machine (SVM)

Part 4 — Generate Predictions and Evaluate the Model

- Generate predictions using the testing dataset
- Calculate model accuracy
- Display the confusion matrix
- Review the classification report

Part 5 — Interpretation

Answer the following questions in your notebook:

1. How accurate was the model?
2. What do the confusion matrix results show?
3. Which features may indicate suspicious behavior?
4. How can machine learning support cybersecurity monitoring and threat detection?

📦 Submission Requirements

Submit the following through Canvas:

✅ Completed Jupyter Notebook (.ipynb)

✅ All notebook cells must be executed and display outputs

✅ Include written responses for the interpretation questions